INCH - POUND

A-A-52561
August 15, 1996

COMMERCIAL ITEM DESCRIPTION

INK, PRINTING: LITHOGRAPHIC OFFSET, WITH DRIER

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-I-3606, for all federal agencies.

1. SCOPE. This commercial item description covers high speed, lithographic, offset printing ink that is compounded with a separate drier, to be mixed together at the time of use. This is to accelerate the drying time and prolong storage life. These inks are formulated for overprinting; intended primarily for field preparation of topographic maps, under varying climatic conditions; and, except for black, are transparent.

2. CLASSIFICATION. The printing ink, with drier, shall conform to the following types and sizes.

2.1 Type.

-1	Black, (Halftone)
-2	Brown, (Contour)
-3	Red
-4	Yellow, (Chrome)
-5	Green, (Dark)
-6	Blue, (Drainage)
-7	Purple, (Offset)

2.2 Size.

-A	1 lb container
-B	5 lb container

Beneficial comments recommendations, additions, deletions, clarification, etc., and any other data which may improve this document should be sent to: U.S. ARMY TOPOGRAPHIC ENGINEERING CENTER, ATTN CETEC-PD-TE, 7701 TELEGRAPH ROAD, ALEXANDRIA, VA 22315-3864.

AMSC N/A FSC 7510

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3. SALIENT CHARACTERISTICS.

3.1 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all the provisions of this specification. Material shall be free from chemicals or compounds that will react adversely with other materials, in the lithographic process, or be injurious to operating personnel.

3.2 Formulation. The ink shall be compounded for lithographic offset printing on sheet-fed presses. The drier shall be separate. The ink shall be formulated to permit its use directly from the container with only the drier being added.

3.2.1 Compatibility. The ink shall be compatible with similar inks when overprinted thereon, and shall be suitable for both line and halftone reproduction.

3.3 Coloration table. The individual item requirements shall be as specified herein and in accordance with the applicable dash letter in table I for the specified color (see 7.4). In the event of conflict between the requirements in the text of this document and the requirements specified in table I, the latter shall govern.

3.4 Certification. When specified in the contract or purchase order (see 7.3), the certification of commercial products as specified in Section 5 may be accepted in lieu of the testing specified herein. When testing is required, these subordinate requirements are mandatory.

3.4.1 Standard sample. Unless otherwise specified in the contract or purchase order (see 7.3), the contractor shall furnish a standard wet sample, a press specimen or a set of drawdowns (see 7.5). When a wet sample is furnished, the contractor shall prepare test drawdowns on the same paper and with the same film applicator as those used to prepare standard drawdowns (see 5.3).

3.4.2 Drawdowns. Test drawdowns for the preparation of color masstone, tinctorial strength, and drying time test shall be made using a 0.0005 Bird film applicator. Test drawdowns for the preparation of color undertone shall be made using a flat, straight-edged instrument not less than 2-inches in width. Drawdowns shall be on paper conforming to The Joint Committee on Printing (JCP) Publication - Designation A60.

3.5 Storage temperature. The packaged ink and drier shall be formulated to withstand prolonged storage at temperatures ranging from -30° ±5° F (-34° ±3° C) to +160° ±5° F (+71° ±3° C). The packaged ink and supplementary drier shall show no signs of gelling, livering, any change in viscosity, or deterioration by storage as specified in 5.3.1.

3.6 Composition.

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3.6.1 Pigments. The pigments used in the ink shall be insoluble in water and dilute acids.

3.6.2 Vehicles. The vehicles used in the ink shall have wetting properties to protect the pigment against chemical action and bleeding. Wetting agents shall not be used that will detrimentally affect the working properties of the ink by introducing emulsification on the press.

3.6.3 Driers. Black ink shall incorporate a minimal amount of cobalt linoleate, factory-mixed into the ink. Transparent ink shall contain no cobalt drier.

3.6.4 Drying characteristics. Unless otherwise specified (see 7.3), the ink without additional drier shall dry in 24 ± 4 hours, in an ambient temperature of + 73° ± 4° F (+ 23 ± 2° C) and a relative humidity of 50% ± 2%. The ink shall dry in not greater than 8 hours, at a temperature of +73° ±4° F and a relative humidity of 50% ± 2% when the supplementary drier, in the amount recommended by the manufacturer, is added to the ink. Drying times (see 7.7) shall be determined as specified in 5.3.2.

3.7 Specific gravity. The specific gravity of the ink shall be not less than 1.10 and not greater than 1.26, at a temperature of +73° ±4° F (+23° ±2° C) when tested in accordance with 5.3.3.

3.8 Tinctorial strength. The tinctorial strength of the ink shall match the tinctorial strength of a standard drawdown when tested as specified in 5.3.4.

3.9 Light fastness. Unless otherwise specified (see 7.3), the light fastness of the ink shall be not less than the light fastness of a standard drawdown when tested as specified in 5.3.5.

3.10 Color. The color of the ink shall be as specified (see 7.3), and shall match the color of the applicable dash number (see table I) when tested as specified in 5.3.6.

3.11 Printing. The ink shall produce clean, open 150- and 300-line shadow dot line copy, a fine 150- and 300-line highlight dot and a solid bar, on paper conforming to JCP Publication - Designation A60, when tested as specified in 5.3.7.

3.12 Identification marking. Each container of ink shall be clearly and legibly lithographed or marked with water-resistant ink. Alternatively, a label printed with water-resistant ink on a good grade of printing paper shall be firmly affixed to the container with a water-resistant adhesive. The markings shall clearly identify the product and its manufacturer.

3.12.1 Additional marking. In addition, the date of manufacture and the recommended amount of drier to be added to obtain a drying time of not greater than 8 hours, at 73° ±3° F (23° ±2° C) with 50% relative humidity, shall also be marked on the container.

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3.13 Metric products. Products manufactured to metric units will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified, and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric units and exceeds the tolerances specified for inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

3.14 Workmanship. Workmanship shall be certified as specified in 5.1.

4 REGULATORY REQUIREMENTS.

4.1 Material Safety Data Sheet. A material safety data sheet shall be prepared in accordance with the requirements of the contract or purchase order, for all chemicals used in the off-set printing process. Special emphasis shall be given to proper disposal of chemical waste.

5.0 QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Market acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID.

a. The company producing the printing ink must have been producing a product meeting the requirements of this CID for at least 2 years.
b. The company must have sold 1000 units meeting this CID in the commercial market place over the past 2 years.

5.3 Tests. When testing is required (see 7.3), all subordinate tests are mandatory. All tests shall be performed on the same run or batch of paper.

5.3.1 Storage temperature tests. Ink and drier samples shall be tested for the ability to withstand prolonged storage at temperature extremes. Any signs of gelling, livering, any change in viscosity, or deterioration observed on examination, after storage as specified, shall constitute failure of these tests.

5.3.1.1 High temperature. The ink and drier shall be placed in a preheated circulating air oven at a temperature of +160° ±5° F (+71° C) for not less than four hours, after which they shall be removed, allowed to cool to +68° ±5° F (20° ±3° C), opened and examined.

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5.3.1.2 Low temperature. The ink and drier shall be placed in a pre-cooled refrigerated chamber at a temperature of -30° F (-34° C) for not less than twelve hours, after which they shall be removed, allowed to warm to +68° F ±5° F (20° C ±3° C), opened and examined.

5.3.2 Drying time test. Maintain an ambient temperature of 73° F ±4° F (+23° C ±2° C), and a relative humidity of 50 % ±2 % during this test. Determine the drying time of the ink by making a drawdown without, and another with the recommended amount of drier. A drying time greater than 24 ±4 hours without drier, or greater than 8 hours with drier, shall be cause for rejection.

5.3.3 Specific gravity test. The specific gravity of the ink shall be tested in accordance with National Printing Ink Research Institute (NPIRI) Standard Test Method D-2 or its equivalent. A specific gravity of less than 1.10 or greater than 1.26 shall be cause for rejection.

5.3.4 Tinctorial strength test. Test the tinctorial strength of the sample drawdowns, using a formulation of bleaching white as follows:

Ingredient	Amount in grams
Zinc oxide (white seal)	79.03 ± 0.05
No. 1 transparent (clear) varnish	19.74 ± 0.02
White petroleum	1.23 ± 0.01.

5.3.4.1 Colors. Mix 5 grams of bleaching white with 100 milligrams (mg) of ink. Prepare a drawdown and immediately compare this drawdown with the standard ink-plus-bleaching white drawdown. Sample drawdowns having a lesser strength than the standard shall constitute failure of this test. Drawdowns of greater strength shall be reduced and retested. Reduce the specimen by adding 50 mg of bleaching white to 1 gram of the sample ink-plus-bleaching white mixture. Prepare a new drawdown and immediately compare the new sample drawdown to the standard. Reduced drawdowns having a greater strength than the standard drawdown shall also constitute failure of this test.

5.3.4.2 Blacks. Mix 5 grams of bleaching white with 100 mg of ink. Prepare a drawdown and immediately compare this drawdown with a standard ink-plus-bleaching white drawdown. Drawdowns having a lesser strength than the standard drawdown shall constitute failure of this test.

5.3.5 Light fastness test. The light fastness of the ink shall be determined using a fadeometer, ASTM G 27, 2500 watt, Method A for 70 hours. Observe and record the results of the test. A light fastness of the undertones of the sample drawdown that is less than that of the standard drawdown shall constitute failure of this test.

5.3.6 Color test. The color of the ink shall be tested against drawdown standards prepared in accordance with table 1, and approved by the government. Drawdowns shall be prepared as

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specified in 3.4.2 and allowed to dry. After the print test specified in 3.11, the drawdowns and press specimens shall be compared in accordance with ASTM D 1729 and D 2244. Comparisons shall be made by persons who have passed a Hardy-Rank Rittler pseudo-isochromatic plate test or equivalent color perception test, or who have been certified by a licensed ophthalmologist as having full-range trichromatic vision. Failure of the drawdown or specimen of the ink to match that made from the Government approved standard shall constitute failure of this test.

5.3.7 Printing test. Run 5,000 impressions using line copy and 150- to 300-line halftone copy and a solid bar on paper conforming to JCP Publication - Designation A60. The run shall be made on a standard, sheet-fed, offset press, 17 by 22 inches or larger, operating at a speed ranging from 4,000 to 6,000 impressions per hour. The plates used shall be aluminum and shall be pre-sensitized either commercially or in accordance with GATF No. 502/4. Examine not less than one of every 100 sheets with a 9-power magnifying glass. Any changes in the size of the dot structure shall constitute failure of this test.

6. PACKAGING.

6.1 Packaging. Preservation, packing and marking shall be as specified in the contract or purchase order (see 7.3).

7. NOTES.

7.1 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

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This example describes a part numbering system for CID A-A-52561:

AA52561-3-A

Size A	(one pound size)
Size B	(five pound size)
-1	(Black, Halftone)
-2	(Brown, Contour)
-3	(Red)
-4	(Yellow, Chrome)
-5	(Green, Dark)
-6	(Blue, Drainage)
-7	(Purple, Offset)
-8	(Gray, Contour)

7.2 Source of documents.

7.2.1 The Joint Committee on Printing (JCP) Publication; Government Paper Specification Standards - Designation A60 is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.2.2 Commercially adopted FED-STD-595 and FED-STD-313 are available from the Naval Publishing and Printing Service Office (NPPSO), Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

7.2.3 ASTM D 1729, ASTM D 2244 and ASTM G 27 are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

7.2.4 NPIRI Standard Test Methods is available from the National Printing Ink Research Institute, Francis MacDonald Sinclair Memorial Laboratory, Lehigh University, Building 7, Bethlehem, PA 18015.

7.2.5 GATF No. 502/4 is available from the Graphic Arts Technical Foundation, 4615 Forbes Avenue, Pittsburgh, PA 15213.

7.3 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Size of container required (see 2.1 and 7.1).
c. Color required (see 3.10).
d. Type of standard sample to be supplied by the government (see 3.4.1).
e. Drying characteristics of ink if other than specified (see 3.6.4).
f. Light fastness required if other than as specified (see 3.9).
g. Color, if a special color is required (see 3.10).

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h. When supplementary drier is not required (see 3.6.4).
i.. When certification may be accepted in lieu of the test (see 3.4).
j. Preservation, packaging, packing, labeling, and marking required.

7.4 National Stock Numbers(NSN). NSNs are available for each dash number, as follows:

-1	Black, Halftone	NSN 7510-00-285-2510
-2	Brown, Contour	NSN 7510-00-224-6719
-3	Red	NSN 7510-00-285-2511
-4	Yellow, Chrome	NSN 7510-00-269-8769
-5	Green, Dark	NSN 7510-00-224-6720
-6	Blue, Drainage	NSN 7510-00-286-4712
-7	Purple, Offset	NSN 7510-00-069-9042
-8	Gray, Contour	NSN 7510-00-034-1470

7.5 Samples. The contracting officer should procure the necessary quantity of standard samples to accompany the solicitation. Samples are to be prepared by the contractor, and shall be in accordance with the requirements of table 1. The contracting officer shall arrange for the inspection and approval of the samples. A set of drawdowns consists of masstone, a tinctorial strength, and an undertone or a press specimen.

7.6 Material Safety Data Sheet. The contracting officer should specify the mailing address for the submittal of The Material Safety Data Sheet as directed in FED-STD-313.

7.7 Drying time. Drying time, is defined as the time lapse between the preparation of the drawdown and when a 3 to 4 pound finger pressure on the surface of the drawdown no longer leaves an impression.

7.7 Subject term (key word) listing.
masstone
tinctorial
spectrophotometric
tristimulus
reflectance

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TABLE I. Coloration parameters by dash number.

DASH NO	-1	-2	-3	-4	-5	-6	-7	-8
COLOR	BLACK NOTE 3	BROWN	RED	YELLOW	GREEN	BLUE	PURPLE	GRAY
COLOR CHIP NOTE 1	58600	58552	60862	57875	52813	47651	95151	57003
ISCC-NBS CENTROID NOTE 2	267	054	012	082	139	178	207	265
SPECTROPHOTOMETRIC	X=9.27 Y=9.27 Z=10.09	X=34.08 Y=30.41 Z=14.26	X=40.39 Y=25.72 Z=11.42	X=73.80 Y=74.13 Z=17.22	Y=27.97 Y=40.17 Z=31.58	X=17.18 Y=16.91 Z=54.77	X=33.65 Y=18.85 Z=55.56	X=38.82 Y=39.80 Z=45.67
TRISTIMULUS DATA	X=0.3238 Y=0.3238	X=0.4328 Y=0.3262	X=0.5210 Y=0.3317	X=0.4469 Y=0.4489	X=0.2805 Y=0.4028	X=0.1933 Y=0.1903	X=0.3114 Y=0.1744	X=0.3123 Y=0.3202
OMINANT WAVELENGTH (MILLIMICRON)	586	585.4	607.7	578.1	528	476.1	550.7	570

NOTE: 1 Color chip number per FED-STD-595.
NOTE: 2 ISCC-NBS CENTROID, is the Interstate Commerce Commission - National Bureau of Standards centroid number.
NOTE: 3 The -1, HALFTONE BLACK, ink is opaque. All the rest are transparent.
NOTE: 4 Wave lengths, (the same for all dash nos.) shall be as follows: 420, 445, 465, 485, 5O5, 525, 545, 570, 595, 615, AND 650.
NOTE: 5 Reflectance, in % of Munsell scale (1 black to 10 white).

-1 HALFTONE BLACK;	8.60, 8.50, 8.55, 8.60, 8.75, 8.85, 9.05, 9.40, 9.70, 9.95, 10.30
-2 CONTOUR BROWN;	9.70, 10.10, 12.30, 17.60, 25.25, 25.00, 23.80, 23.10, 46.70, 52.15, 54.10
-3 RED;	10.80, 10.45, 9.05, 7.80, 7.90, 8.85, 10.80, 15.30, 52.50, 80.25, 85.70
-4 CHROME YELLOW;	10.80, 11.40, 13.45, 20.00, 44.95, 61.40, 68.20, 87.00, 96.95, 98.25, 98.60
-5 DARK GREEN;	24.20, 23.65, 25.25, 32.10, 58.45, 64.40, 49.65, 30.40, 24.95, 23.70, 26.60
-6 DRAINAGE BLUE;	39.40, 47.75, 50.35, 46.30, 37.00, 24.60, 14.05, 8.20, 8.35, 9.20, 13.30
-7 OFFSET PURPLE;	44.60, 55.25, 49.60, 30.50, 15.50, 9.40, 9.05, 9.80, 16.25, 43.35, 84.60
-8 CONTOUR GRAY;	37.30, 38.40, 39.20, 39.85, 40.35, 40.25, 39.85, 39.20, 39.40, 39.95, 41.90

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Review activity:
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GSA - FSS

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